

Circuit Protection for AC LED Lighting Systems

Pairing a polymeric positive temperature coefficient device with a metal-oxide varistor provides a completely resettable and coordinated circuit protection solution for AC LED lighting systems



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In recent years, light-emitting diode (LED) technology has made impressive gains in price, performance and manufacturability. Further improvements that optimise LED operation are expected to drive exponential growth in the LED lighting market. One of the most recent commercial developments is a family of high-voltage LEDs (AC LEDs) that can operate directly off line voltage without the need for an AC-DC converter.

AC LED lighting systems offer a high overall system reliability. However, as in any LED lighting application, protection against overheating caused by transients is an important design consideration. In this article,

circuit protection strategies that can help prevent overheating in AC LED lighting are discussed, and measured results in a typical circuit included.

Protection devices used

As with any electronics system operating off line voltage, without adequate protection, AC LED lighting systems can be damaged by lightning surges or ring waves. A metal-oxide varistor (MOV) is often used to help protect AC LED lighting systems in such cases. It helps manufacturers meet safety and performance standards. Lightning tests according to IEC 61000-4-5 and ring wave tests according to IEEE C.62.41 can be used to simulate these real-life threats in the lab.

An AC LED, like a DC LED, is a semiconductor device that converts electrical energy into light when cur-

rent passes through the LED's die. Excessive heat at the junction can dramatically reduce both the light output and the lifespan of the LED. When proper thermal management and circuit protection design strategies are employed, AC LED technology offers several advantages over conventional lighting. These include compact size, component count reduction, energy efficiency and reduced system cost.

Polymeric positive temperature coefficient (PPTC) devices help provide over-current and over-temperature protection and can be easily integrated onto the circuit board with the AC LED.

Over-current and thermal runaway design considerations

LEDs exhibit a negative thermal coefficient. LED's forward voltage (V_f)

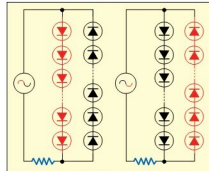


Fig. 1: Typical AC LED schematic

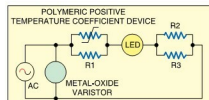


Fig. 2: AC LED AN3211 test set-up